

AVD Manual

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1. Architecture & Responsibility (Visibility)

To understand what is happening in your environment, you often need to query both the Control Plane (Microsoft's side) and the Data Plane (Your VMs).

PowerShell

Query the Control Plane (Host Pools)

```
Get-AzWvdHostPool -ResourceGroupName "RG-AVD-Prod" | Select-Object Name, HostPoolType, LoadBalancerType
```

Query the Data Plane (Your actual Virtual Machines)

```
Get-AzVM -ResourceGroupName "RG-AVD-Prod" | Select-Object Name, ProvisioningState
```

2. Core Prerequisites (Validation)

Before deploying or troubleshooting a session host, verify it meets the foundational identity and networking requirements.

PowerShell

1. Verify Active Directory Domain Trust is intact

```
Test-ComputerSecureChannel -Verbose
```

2. Test outbound connectivity to the primary AVD Broker

```
Test-NetConnection -ComputerName "rdbroker.wvd.microsoft.com" -Port 443
```

3. Session Host Sizing (Auditing)

Quickly audit your environment to see if your VM SKUs align with the Light/Medium/Heavy workload recommendations.

PowerShell

List all AVD Session Hosts and their exact hardware sizes

```
Get-AzVM -ResourceGroupName "RG-AVD-Prod" |  
    Select-Object Name, @{Name="VM Size";Expression={$_.HardwareProfile.VmSize}} |  
    Sort-Object "VM Size"
```

4. Essential AVD PowerShell Commands (Session Management)

Finding and managing stranded or disconnected users is a daily administrative task.

PowerShell

Find all Disconnected sessions in a Host Pool

```
$sessions = Get-AzWvdUserSession -ResourceGroupName "RG-AVD-Prod"
-HostPoolName "HP-Finance" |
  Where-Object SessionState -eq "Disconnected"
```

Force logoff all disconnected sessions

```
$sessions | ForEach-Object {
    Remove-AzWvdUserSession -ResourceGroupName "RG-AVD-Prod" -HostPoolName
    "HP-Finance" -SessionHostName $_.SessionHostName -Id $_.Id -Force
}
```

5. FSLogix Profile Management (Deployment)

Automate the configuration of the "Gold Standard" FSLogix settings on your golden image or via a startup script.

PowerShell

Configure Core FSLogix Settings via Registry

```
$fslogixPath = "HKLM:\SOFTWARE\FSLogix\Profiles"
```

```
if (!(Test-Path $fslogixPath)) { New-Item -Path $fslogixPath -Force | Out-Null }
```

```
$settings = @{
```

```
    "Enabled" = 1
```

```
    "VHDLocations" = "\\YourStorageAccount.file.core.windows.net\profiles"
```

```
    "VolumeType" = "vhdx"
```

```
    "DeleteLocalProfileWhenVHDSshouldApply" = 1
```

```
    "FlipFlopProfileDirectoryName" = 1
```

```
}
```

```
foreach ($key in $settings.Keys) {
```

```
    Set-ItemProperty -Path $fslogixPath -Name $key -Value $settings[$key] -Type
```

```
$(if($settings[$key] -is [int]){DWord}else{String}) -Force
```

```
}
```

6. Critical Optimizations & Features (RDP Shortpath)

Enable RDP Shortpath for Managed Networks to bypass the AVD gateway and drastically improve user latency.

PowerShell

Enable UDP Port Redirection for RDP Shortpath

\$rdpPath = "HKLM:\SYSTEM\CurrentControlSet\Control\Terminal Server\WinStations"

Set-ItemProperty -Path \$rdpPath -Name "fUseUdpPortRedirector" -Value 1 -Type DWord -Force

Write-Host "RDP Shortpath enabled. Restart the 'Remote Desktop Services' (TermService) to apply." -ForegroundColor Green

7. Troubleshooting Quick Reference (Diagnostics)

When a user calls in with a black screen or login failure, use this script on the target Session Host to immediately check the agent and profile logs.

PowerShell

1. Verify AVD Infrastructure Agents are running

Get-Service -Name "RDAgentBootLoader", "RemoteAccess" | Select-Object Name, Status

2. Tail the latest FSLogix Profile log for errors (Reads the last 15 lines)

\$latestLog = Get-ChildItem -Path "C:\ProgramData\FSLogix\Logs\Profile*.log" | Sort-Object LastWriteTime -Descending | Select-Object -First 1

if (\$latestLog) {

 Write-Host "--- Latest FSLogix Log: \$(\$latestLog.Name) ---" -ForegroundColor Cyan

 Get-Content -Path \$latestLog.FullName -Tail 15

}

8. Essential AVD PowerShell Commands

Prerequisite: Install-Module -Name Az.DesktopVirtualization

Authenticate to Azure

Connect-AzAccount

Get all Host Pools in a Resource Group

Get-AzWvdHostPool -ResourceGroupName "RG-AVD-Prod"

Put a Session Host into Drain Mode (Prevent new logins, allow existing to finish)

Update-AzWvdSessionHost -ResourceGroupName "RG-AVD-Prod" `

 -HostPoolName "HP-Finance" `

 -Name "HP-Finance-0.contoso.com" `

 -AllowNewSession:\$false

```
# Log off a specific user session forcefully
Remove-AzWvdUserSession -ResourceGroupName "RG-AVD-Prod" `
  -HostPoolName "HP-Finance" `
  -SessionHostName "HP-Finance-0.contoso.com" `
  -Id 3 # Replace with actual Session ID

# Generate a new Registration Token (Valid for 2 hours)
$ExpirationTime = (Get-Date).AddHours(2).ToString('yyyy-MM-ddTHH:mm:ssZ')
New-AzWvdRegistrationInfo -ResourceGroupName "RG-AVD-Prod" `
  -HostPoolName "HP-Finance" `
  -ExpirationTime $ExpirationTime
```

6. Critical Optimizations & Features

- **RDP Shortpath:** Establishes a direct UDP connection between the Remote Desktop Client and the Session Host, bypassing the AVD Gateway for vastly improved latency and bandwidth. Requires UDP traffic allowed on firewalls.
- **Start VM on Connect:** Reduces cloud costs. VMs remain deallocated until a user clicks their icon in the AVD client, prompting Azure to boot the VM.
- **Teams Audio/Video Redirection:** Install the WebRTC Redirector Service on the golden image. Shifts media processing from the cloud VM to the user's local endpoint.
- **Multimedia Redirection (MMR):** Smooths out video playback in Edge/Chrome by rendering the video on the local endpoint instead of the session host.

7. Troubleshooting Quick Reference

- **Black Screen on Login:** Often an FSLogix issue. Check the C:\ProgramData\FSLogix\Logs\Profile directory on the session host.
- **VM Shows "Unavailable":** The AVD Agent on the VM cannot reach the broker. Restart the Remote Desktop Services Infrastructure Agent (RDAgentBootLoader) service, or verify the VM has outbound internet access.
- **User "No Resources Assigned":** Verify the user is assigned to an Application Group, and that the App Group is assigned to a Workspace. Check Entra ID group sync if using Hybrid identity.

8. Architecture & Responsibility (Control and Data Plane)

AVD divides management into two planes: what Microsoft handles (Control Plane) and what you manage (Data Plane).

Microsoft Manages (Control Plane)	You Manage (Customer Data Plane)
Web Access, Gateway, Broker, Diagnostics	Azure Subscriptions, Virtual Network (VNet)
Load Balancing (Breadth/Depth)	Azure AD / Active Directory Identity
Global Azure Infrastructure	Session Hosts (VMs), OS Images, Apps
Extensibility (REST APIs)	User Profiles (FSLogix) & Policies

9. Session Host Sizing Guidelines (Windows 11 Multi-Session)

Based on vCPU to User ratios.

Workload Type	Max Users per vCPU	Recommended VM Series	Example Use Case
Light	6	D-Series (D8s_v5)	Data entry, lightweight web apps
Medium	4	D-Series (D8s_v5)	Word, Excel, web browsing
Heavy	2	D-Series, E-Series	Outlook, Teams (AV optimized), heavy Excel

Power	1	N-Series (NVv4/NVv5)	CAD, 3D rendering, video editing
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